Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

by: blanketism

began: July 4, 2026
last edited: August 5, 2026

Contents

5D Diagonalizable Operators

Exercise 5D1

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$.

(a) Prove that if $T^4 = I$, then T is diagonalizable.

(b) Prove that if $T^4 = T$, then T is diagonalizable.

(c) Give an example of an operator $T \in \mathcal{L}(\mathbb{C}^2)$ such that $T^4 = T^2$ and T is not diagonalizable.

Manual Remark. *The roots ω in part (b) are the roots obtained in* Exercise 1C6.

Solution.

(a) If $T^4 = I$, then the minimal polynomial of T divides $z^4 - 1 = (z-1)(z+1)(z-i)(z+i)$, which has no repeated roots. Hence the minimal polynomial of T has no repeated roots, so T is diagonalizable.

(b) If $T^4 = T$, then the minimal polynomial of T divides $z^4 - z = z(z-1)(z-\omega)(z-\omega^2)$. This polynomial has no repeated roots, so the minimal polynomial of T has no repeated roots, and hence T is diagonalizable.

(c) Let $T \in \mathcal{L}(\mathbb{C}^2)$ be defined by

$$T(z_1, z_2) = (z_2, 0).$$

Then $T^2(z_1, z_2) = (0, 0)$, so $T^4 = T^2$. However, T is not diagonalizable because the only eigenvalue of T is 0, and $\dim E(0, T) = 1 < 2 = \dim \mathbb{C}^2$. ■

Exercise 5D2

Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix A with respect to some basis of V. Prove that if $\lambda \in \mathbb{F}$, then λ appears on the diagonal of A precisely $\dim E(\lambda, T)$ times.

Solution. Suppose A is a diagonal matrix of T with respect to some basis of V. Let $\lambda \in \mathbb{F}$. Let m denote the number of times λ appears on the diagonal of A. Then $A - \lambda I$ is a diagonal matrix of $T - \lambda I$ with respect to the same basis, with exactly m zeros on the diagonal. A diagonal matrix with m zeros on the diagonal has null space of dimension m, so $\dim \operatorname{null}(T - \lambda I) = m$. Since $E(\lambda, T) = \operatorname{null}(T - \lambda I)$, we have $\dim E(\lambda, T) = m$. ■

Exercise 5D3

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that if the operator T is diagonalizable, then $V = \operatorname{null} T \oplus \operatorname{range} T$.

Solution. Suppose T is diagonalizable. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Then

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T).$$

If 0 is not an eigenvalue of T, then $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = V$, so $V = \operatorname{null} T \oplus \operatorname{range} T$. If 0 is an eigenvalue of T, then we may assume $\lambda_1 = 0$. Let

$$U = E(\lambda_2, T) \oplus \cdots \oplus E(\lambda_m, T).$$

Since $\operatorname{null} T = E(0, T)$, we have $V = \operatorname{null} T \oplus U$. It remains to show that $U = \operatorname{range} T$.

Let $u \in U$. Then $u = u_2 + \cdots + u_m$ for some $u_k \in E(\lambda_k, T)$. Since $\lambda_k \neq 0$ for $k = 2, \ldots, m$, we have $u_k = \lambda_k^{-1} T(u_k)$ for each $k = 2, \ldots, m$. Hence,

$$u = u_2 + \cdots + u_m = \lambda_2^{-1} T(u_2) + \cdots + \lambda_m^{-1} T(u_m) = T(\lambda_2^{-1} u_2 + \cdots + \lambda_m^{-1} u_m) \in \operatorname{range} T.$$

Thus, $U \subseteq \operatorname{range} T$.

Now let $v \in \operatorname{range} T$. Note that eigenspaces are invariant as follows: If $u \in E(\lambda_k, T)$, then $T(u) = \lambda_k u \in E(\lambda_k, T)$. Since U is a direct sum of eigenspaces, it is invariant under T. Then $v = T(w)$ for some $w \in V$. Since $V = \operatorname{null} T \oplus U$, we may write $w = w_1 + w_2$ for some $w_1 \in \operatorname{null} T$ and $w_2 \in U$. Then

$$v = T(w) = T(w_1 + w_2) = T(w_1) + T(w_2) = 0 + T(w_2) = T(w_2) \in U.$$

Thus, $\operatorname{range} T \subseteq U$. Since $U \subseteq \operatorname{range} T$ and $\operatorname{range} T \subseteq U$, we have $U = \operatorname{range} T$. Therefore, $V = \operatorname{null} T \oplus U = \operatorname{null} T \oplus \operatorname{range} T$. ∎

Exercise 5D4

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $V = \operatorname{null} T \oplus \operatorname{range} T$.
(b) $V = \operatorname{null} T + \operatorname{range} T$.
(c) $\operatorname{null} T \cap \operatorname{range} T = \{0\}$.

Solution.

(a) If $V = \operatorname{null} T \oplus \operatorname{range} T$, then $V = \operatorname{null} T + \operatorname{range} T$.
(b) If $V = \operatorname{null} T + \operatorname{range} T$, then $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T - \dim(\operatorname{null} T \cap \operatorname{range} T)$. Since $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ by the fundamental theorem of linear maps, we have $\dim(\operatorname{null} T \cap \operatorname{range} T) = 0$, so $\operatorname{null} T \cap \operatorname{range} T = \{0\}$.
(c) If $\operatorname{null} T \cap \operatorname{range} T = \{0\}$, then $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T - \dim(\operatorname{null} T \cap \operatorname{range} T) = \dim \operatorname{null} T + \dim \operatorname{range} T$. Since $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ by the fundamental theorem of linear maps, and $\operatorname{null} T \cap \operatorname{range} T = \{0\}$, we have $V = \operatorname{null} T \oplus \operatorname{range} T$. ∎

Exercise 5D5

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I)$$

for every $\lambda \in \mathbb{C}$.

Solution. Suppose T is diagonalizable. Let $\lambda \in \mathbb{C}$. Since T is diagonalizable, we have

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$$

for some distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ of T. If λ is not an eigenvalue of T, then $\operatorname{null}(T - \lambda I) = \{0\}$ and $\operatorname{range}(T - \lambda I) = V$, so

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I).$$

If λ is an eigenvalue of T, then we may assume $\lambda_1 = \lambda$. Let

$$U = E(\lambda_2, T) \oplus \cdots \oplus E(\lambda_m, T).$$

Then $V = E(\lambda, T) \oplus U$. It remains to show that $U = \operatorname{range}(T - \lambda I)$.

By dimension counting, we have

$$\dim U = \dim V - \dim E(\lambda, T) = \dim V - \dim \operatorname{null}(T - \lambda I) = \dim \operatorname{range}(T - \lambda I).$$

Since each $u_k \in E(\lambda_k, T)$ satisfies $T(u_k) = \lambda_k u_k$, we have

$$(T - \lambda I)(u_k) = (\lambda_k - \lambda)u_k \in E(\lambda_k, T).$$

To see that $T - \lambda I$ is injective on U, suppose $u \in U$ and $(T - \lambda I)(u) = 0$. Then $u = u_2 + \cdots + u_m$ for some $u_k \in E(\lambda_k, T)$, and

$$0 = (T-\lambda I)(u) = (T-\lambda I)(u_2+\cdots+u_m) = (T-\lambda I)(u_2)+\cdots+(T-\lambda I)(u_m) = (\lambda_2-\lambda)u_2+\cdots+(\lambda_m-\lambda)u_m.$$

Since $\lambda_2, \ldots, \lambda_m$ are distinct from λ, we have $\lambda_k - \lambda \neq 0$ for each $k = 2, \ldots, m$. Since the sum above is a direct sum, we have $(\lambda_k - \lambda)u_k = 0$ for each $k = 2, \ldots, m$, so $u_k = 0$ for each $k = 2, \ldots, m$. Hence, $u = u_2 + \cdots + u_m = 0$, so $T - \lambda I$ is injective on U. Since $T - \lambda I$ is injective on U and $\dim U = \dim \operatorname{range}(T - \lambda I)$, we have $U = \operatorname{range}(T - \lambda I)$. ■

Exercise 5D6

Suppose $T \in \mathcal{L}(\mathbb{F}^5)$ and $\dim E(8, T) = 4$. Prove that $T - 2I$ or $T - 6I$ is invertible.

Solution. Suppose $T - 2I$ and $T - 6I$ are not invertible. Then 2 and 6 are eigenvalues of T, so $\dim E(2, T) \geq 1$ and $\dim E(6, T) \geq 1$. Since 8 is also an eigenvalue of T, we have

$$\dim E(2, T) + \dim E(6, T) + \dim E(8, T) \geq 1 + 1 + 4 = 6 > 5 = \dim \mathbb{F}^5,$$

which is a contradiction. Hence, $T - 2I$ or $T - 6I$ is invertible. ■

Exercise 5D7

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$E(\lambda, T) = E\left(\frac{1}{\lambda}, T^{-1}\right)$$

for every $\lambda \in \mathbb{F}$ with $\lambda \neq 0$.

Solution. Suppose $v \in E(\lambda, T)$. Then $T(v) = \lambda v$. Since T is invertible, we have $v = T^{-1}(\lambda v) = \lambda T^{-1}(v)$, so $T^{-1}(v) = \frac{1}{\lambda}v$. Hence,

$$v \in E(\lambda^{-1}, T^{-1}),$$

so $E(\lambda, T) \subseteq E(\lambda^{-1}, T^{-1})$.

Now suppose $v \in E(\lambda^{-1}, T^{-1})$. Then

$$T^{-1}(v) = \lambda^{-1}v \implies v = T(\lambda^{-1}v) \implies v = \lambda^{-1}T(v) \implies T(v) = \lambda v.$$

Hence, $v \in E(\lambda, T)$, so $E(\lambda^{-1}, T^{-1}) \subseteq E(\lambda, T)$. Then $E(\lambda, T) = E(\lambda^{-1}, T^{-1})$. ■

Exercise 5D8

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct nonzero eigenvalues of T. Prove that

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim \operatorname{range} T.$$

Solution. Let $U = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$. Then $\dim U = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$. To see that T is injective on U, suppose $u \in U$ and $T(u) = 0$. Then $u = u_1 + \cdots + u_m$ for some $u_k \in E(\lambda_k, T)$, and

$$0 = T(u) = T(u_1 + \cdots + u_m) = T(u_1) + \cdots + T(u_m) = \lambda_1 u_1 + \cdots + \lambda_m u_m.$$

Since $\lambda_1, \ldots, \lambda_m$ are distinct and nonzero, we have $\lambda_k \neq 0$ for each $k = 1, \ldots, m$. Since the sum above is a direct sum, we have $\lambda_k u_k = 0$ for each $k = 1, \ldots, m$, so $u_k = 0$ for each $k = 1, \ldots, m$. Hence,

$$\dim U = \dim T(U) \leq \dim \operatorname{range} T.$$

Therefore,

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) = \dim U \leq \dim \operatorname{range} T. \qquad \blacksquare$$

Exercise 5D9

Suppose $R, T \in \mathcal{L}(\mathbb{F}^3)$ each have $2, 6, 7$ as eigenvalues. Prove that there exists an invertible operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $R = S^{-1}TS$.

Solution. Let r_2, r_6, r_7 denote the eigenvectors of R corresponding to the eigenvalues $2, 6, 7$, respectively. Since $\dim V = 3$ and R has three distinct eigenvalues, we have

$$\dim V = \dim E(2, R) + \dim E(6, R) + \dim E(7, R) = 3.$$

Since $\dim E(\lambda, R) \geq 1$ for each eigenvalue $\lambda = 2, 6, 7$, we have $\dim E(\lambda, R) = 1$ for each eigenvalue $\lambda = 2, 6, 7$. Hence, $\{r_2, r_6, r_7\}$ is a basis of V. Let t_2, t_6, t_7 denote the eigenvectors of T corresponding to the eigenvalues $2, 6, 7$, respectively. By the same reasoning, $\{t_2, t_6, t_7\}$ is a basis of V. Let $S \in \mathcal{L}(V)$ be defined by

$$S(r_2) = t_2, \quad S(r_6) = t_6, \quad S(r_7) = t_7.$$

Then S is invertible because it maps a basis of V to a basis of V. To see that $R = S^{-1}TS$, note that for any basis vector r_k of R, we have

$$S^{-1}TS(r_k) = S^{-1}T(t_k) = S^{-1}(\lambda_k t_k) = \lambda_k S^{-1}(t_k) = \lambda_k r_k = R(r_k),$$

where λ_k is the eigenvalue corresponding to r_k and t_k. Since $S^{-1}TS$ and R agree on a basis of V, we have $R = S^{-1}TS$. $\blacksquare$

Exercise 5D10

Find $R, T \in \mathcal{L}(\mathbb{F}^4)$ such that R and T each have $2, 6, 7$ as eigenvalues, R and T have no other eigenvalues, and there does not exist an invertible operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $R = S^{-1}TS$.

Solution. Define R to be the operator whose matrix with respect to the standard basis of $\mathbb{F}^4$ has diagonal entries $2,6,7,7$ and all other entries 0. Then R has eigenvalues $2,6,7$ and no other eigenvalues. Define T to be the operator whose matrix with respect to the standard basis of $\mathbb{F}^4$ has diagonal entries $2,6,7,2$ and all other entries 0. Then T has eigenvalues $2,6,7$ and no other eigenvalues. If there did exist an invertible operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $R = S^{-1}TS$, then

$$Rv = S^{-1}TSv = \lambda v \implies T(Sv) = \lambda Sv$$

for every eigenvalue λ of R and some eigenvector $v \in E(\lambda, R)$. Hence, the map $S|_{E(\lambda,R)}$ maps $E(\lambda, R)$ to $E(\lambda, T)$. Since S is invertible, $S|_{E(\lambda,R)}$ is injective, so $\dim E(\lambda, R) \leq \dim E(\lambda, T)$. However, $\dim E(7,R) = 2 > 1 = \dim E(7,T)$, so there does not exist an invertible operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $R = S^{-1}TS$. ■

Exercise 5D11

Find $T \in \mathcal{L}(\mathbb{C}^3)$ such that 6 and 7 are eigenvalues of T and such that T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$.

Solution. Consider the operator $T \in \mathcal{L}(\mathbb{C}^3)$ whose matrix with respect to the standard basis of $\mathbb{C}^3$ is

$$\begin{pmatrix} 6 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

Observe that 6 and 7 are eigenvalues of T. If T had a diagonal matrix with respect to some basis of $\mathbb{C}^3$, then $\dim E(6,T) + \dim E(7,T) = 3$. However, $E(6,T)$ is the null space of $T - 6I$, which has matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Observe that the null space of this matrix is spanned by $(1,0,0)$, so $\dim E(6,T) = 1$. Similarly, we can deduce that $\dim E(7,T) = 1$. Hence, $\dim E(6,T) + \dim E(7,T) = 2 < 3$, so T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$. ■

Exercise 5D12

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is such that 6 and 7 are eigenvalues of T. Furthermore, suppose T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$. Prove that there exists $(z_1, z_2, z_3) \in \mathbb{C}^3$ such that

$$T(z_1, z_2, z_3) = (6 + 8z_1, 7 + 8z_2, 13 + 8z_3).$$

Solution. To show that some $(z_1, z_2, z_3) \in \mathbb{C}^3$ satisfies the above is the same as showing that the operator $T - 8I$ is surjective. Since T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$, we have $\dim E(6,T) + \dim E(7,T) < 3$. We claim that 6 and 7 are the only eigenvalues of T. If there were another eigenvalue λ of T, then $\dim E(6,T) + \dim E(7,T) + \dim E(\lambda,T) \geq 1 + 1 + 1 = 3$, which is a contradiction. Hence, 6 and 7 are the only eigenvalues of T.

Then 8 is not an eigenvalue of T, so $T - 8I$ is injective. Since $T - 8I$ is a linear map from $\mathbb{C}^3$ to $\mathbb{C}^3$, it is injective if and only if it is surjective. Hence, $T - 8I$ is surjective, so there exists $(z_1, z_2, z_3) \in \mathbb{C}^3$ such that

$$(T - 8I)(z_1, z_2, z_3) = (6, 7, 13) \implies T(z_1, z_2, z_3) = (6 + 8z_1, 7 + 8z_2, 13 + 8z_3). \qquad \blacksquare$$

Exercise 5D13

Suppose A is a diagonal matrix with distinct entries on the diagonal and B is a matrix of the same size as A. Show that $AB = BA$ if and only if B is a diagonal matrix.

Solution. Let A be a diagonal matrix with distinct entries on the diagonal, and let B be a matrix of the same size as A. Suppose $AB = BA$. Let $A_{j,j}$ denote the j-th diagonal entry of A, and let $B_{j,k}$ denote the entry of B in row j and column k. Then

$$(AB)_{j,k} = A_{j,j} B_{j,k}, \quad (BA)_{j,k} = B_{j,k} A_{k,k}.$$

Since $AB = BA$, we have

$$A_{j,j} B_{j,k} = B_{j,k} A_{k,k}.$$

If $j \neq k$, then $A_{j,j} - A_{k,k} \neq 0$ because the diagonal entries of A are distinct. Hence,

$$(A_{j,j} - A_{k,k}) B_{j,k} = 0 \implies B_{j,k} = 0.$$

Thus, all off-diagonal entries of B are zero, so B is a diagonal matrix.

Conversely, if B is a diagonal matrix, then it is straightforward to verify that $AB = BA$. $\blacksquare$

Exercise 5D14

(a) Give an example of a finite-dimensional complex vector space and an operator T on that vector space such that T^2 is diagonalizable but T is not diagonalizable.

(b) Suppose $\mathbb{F} = \mathbb{C}$, k is a positive integer, and $T \in \mathcal{L}(V)$ is invertible. Prove that T is diagonalizable if and only if T^k is diagonalizable.

Solution.

(a) Consider the operator $T \in \mathcal{L}(\mathbb{C}^2)$ defined by

$$T(z_1, z_2) = (z_2, 0).$$

Then $T^2(z_1, z_2) = (0, 0)$, so T^2 is diagonalizable. However, T is not diagonalizable because the only eigenvalue of T is 0, and $\dim E(0, T) = 1 < 2 = \dim \mathbb{C}^2$.

(b) Suppose T is diagonalizable. Then a basis of V consists of eigenvectors of T, and each eigenvector of T is also an eigenvector of T^k. Hence, T^k is diagonalizable.

Conversely, suppose T^k is diagonalizable. Then a basis of V consists of eigenvectors of T^k. Consider the minimal polynomial p of T^k as

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_m),$$

where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T^k. Consider the polynomial

$$q(z) = (z^k - \lambda_1) \cdots (z^k - \lambda_m).$$

Note that $q(T) = p(T^k) = 0$, so the minimal polynomial of T divides q. Since T is invertible, 0 is not an eigenvalue of T, so 0 is not a root of the minimal polynomial of T. To see that q has no repeated roots, note that individual $(z^k - \lambda_j)$ factors as k distinct linear factors over $\mathbb{C}$. Moreover, factors from $(z^k - \lambda_j)$ and $(z^k - \lambda_\ell)$ do not share any roots for $j \neq \ell$, for otherwise if ω is a root of both $z^k - \lambda_j$ and $z^k - \lambda_\ell$, then $\omega^k = \lambda_j = \lambda_\ell$, which is a contradiction. Hence, q has no repeated roots, so the minimal polynomial of T has no repeated roots. Since the minimal polynomial of T has no repeated roots, T is diagonalizable. ■

Exercise 5D15

Suppose V is a finite-dimensional complex vector space, $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T. Prove that the following are equivalent.

(a) T is diagonalizable.

(b) There does not exist $\lambda \in \mathbb{C}$ such that p is a polynomial multiple of $(z - \lambda)^2$.

(c) p and its derivative p' have no zeros in common.

(d) The greatest common divisor of p and p' is the constant polynomial 1.

Axler Note. *The* **greatest common divisor** *of p and p' is the monic polynomial q of largest degree such that p and p' are both polynomial multiples of q. The Euclidean algorithm for polynomials (look it up) can quickly determine the greatest common divisor of two polynomials, without requiring any information about the zeros of the polynomials. Thus the equivalence of (a) and (d) above shows that we can determine whether T is diagonalizable without knowing anything about the zeros of p.*

Solution.

(a) $\Longrightarrow$ (b) Suppose T is diagonalizable. Then the minimal polynomial of T has no repeated roots, so there does not exist $\lambda \in \mathbb{C}$ such that p is a polynomial multiple of $(z - \lambda)^2$.

(b) $\Longrightarrow$ (c) Suppose there does not exist $\lambda \in \mathbb{C}$ such that p is a polynomial multiple of $(z - \lambda)^2$. Then p has no repeated roots, so we may let

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_m)$$

for some distinct $\lambda_1, \ldots, \lambda_m \in \mathbb{C}$. Then evaluating the derivative p' at λ_k gives

$$p'(\lambda_k) = \prod_{j \neq k} (\lambda_k - \lambda_j) \neq 0$$

by the distinctness of $\lambda_1, \ldots, \lambda_m$. Hence, p and p' have no zeros in common.

(c) $\Longrightarrow$ (d) Suppose p and its derivative p' have no zeros in common. Then the greatest common divisor of p and p' has no zeros, so it is a constant polynomial.

(d) $\Longrightarrow$ (a) Suppose the greatest common divisor of p and p' is the constant polynomial 1. If p had a repeated root $\lambda \in \mathbb{C}$, then p would be a polynomial multiple of $(z - \lambda)^2$, so p' would be a polynomial multiple of $(z - \lambda)$. Hence, p and p' would have a zero in common, which is a contradiction. Thus, p has no repeated roots, so T is diagonalizable. ■

Exercise 5D16

Suppose that $T \in \mathcal{L}(V)$ is diagonalizable. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Prove that a subspace U of V is invariant under T if and only if there exist subspaces $U_1, \ldots, U_m$ of V such that $U_k \subseteq E(\lambda_k, T)$ for each k and $U = U_1 \oplus \cdots \oplus U_m$.

Solution. Suppose U is invariant under T. Let $U_k = U \cap E(\lambda_k, T)$ for each $k = 1, \ldots, m$. Then $U_k \subseteq E(\lambda_k, T)$ for each k. For each k, define the polynomial p_k by

$$p_k(z) = \prod_{j \neq k} \frac{z - \lambda_j}{\lambda_k - \lambda_j}.$$

Then $p_k(\lambda_j) = 0$ for $j \neq k$ and $p_k(\lambda_k) = 1$. Let $u \in U$ with

$$u = u_1 + \cdots + u_m, \quad u_k \in E(\lambda_k, T).$$

Then $p_k(T)(u) = u_k \in U$ for each k, so $u_k \in U_k$ for each k. Hence, $U = U_1 \oplus \cdots \oplus U_m$.

Conversely, suppose there exist subspaces $U_1, \ldots, U_m$ of V such that $U_k \subseteq E(\lambda_k, T)$ for each k and $U = U_1 \oplus \cdots \oplus U_m$. Then for each $u \in U$, we have

$$u = u_1 + \cdots + u_m, \quad u_k \in U_k.$$

Then

$$T(u) = T(u_1 + \cdots + u_m) = T(u_1) + \cdots + T(u_m) = \lambda_1 u_1 + \cdots + \lambda_m u_m \in U,$$

so U is invariant under T. ■

Exercise 5D17

Suppose V is finite-dimensional. Prove that $\mathcal{L}(V)$ has a basis consisting of diagonalizable operators.

Solution. Let $n = \dim V$. Let $\{v_1, \ldots, v_n\}$ be a basis of V. For each $j, k \in \{1, \ldots, n\}$, define $T_{j,k} \in \mathcal{L}(V)$ by

$$T_{j,k}(v_\ell) = \begin{cases} v_j & \text{if } \ell = k, \\ 0 & \text{if } \ell \neq k. \end{cases}$$

If $j = k$, let $D_j = T_{j,j}$. Then D_j is diagonalizable because $D_j(v_j) = v_j$ and $D_j(v_\ell) = 0$ for $\ell \neq j$. If $j \neq k$, let $N_{j,k} = D_j + T_{j,k}$. Then $N_{j,k}$ is diagonalizable because $N_{j,k}(v_j) = v_j$, $N_{j,k}(v_k) = v_j$, and $N_{j,k}(v_\ell) = 0$ for $\ell \neq j, k$. Then the eigenvalues of $N_{j,k}$ are 0 and 1 with corresponding eigenvectors $v_j - v_k$ ($n-1$ of these) and v_j (1 of these), so $N_{j,k}$ is diagonalizable.

Note that the set

$$\{D_1, \ldots, D_n\} \cup \{N_{j,k} : j, k \in \{1, \ldots, n\}, j \neq k\}$$

spans $\mathcal{L}(V)$ because for any $T \in \mathcal{L}(V)$, we have

$$T = \sum_{j,k=1}^{n} \alpha_{j,k} T_{j,k} = \sum_{j \neq k} \alpha_{j,k}(N_{j,k} - D_j) + \sum_{j=1}^{n} \alpha_{j,j} D_j$$

shows that the set above spans $\mathcal{L}(V)$.

Moreover, note that $\dim \mathcal{L}(V) = n^2$ and the set above has $n + n(n-1) = n^2$ elements, so it is a basis of $\mathcal{L}(V)$ consisting of diagonalizable operators. ■

Exercise 5D18

Suppose that $T \in \mathcal{L}(V)$ is diagonalizable and U is a subspace of V that is invariant under T. Prove that the quotient operator T/U is a diagonalizable operator on V/U.

Solution. Since T is diagonalizable, there exists a basis of V consisting of eigenvectors of T. To see that U has a basis consisting of eigenvectors of T, note that V has a basis of eigenvectors of T. By invariance of U under T, we may consider $T|_U \in \mathcal{L}(U)$ which is also diagonalizable. Hence, U has a basis consisting of eigenvectors of T.

Let $\{v_1, \ldots, v_m\}$ be a basis of U consisting of eigenvectors of T. Extend this basis to a basis $\{v_1, \ldots, v_n\}$ of V consisting of eigenvectors of T. Then $\{v_{m+1} + U, \ldots, v_n + U\}$ is a basis of V/U. For each $k = m+1, \ldots, n$, we have

$$(T/U)(v_k + U) = T(v_k) + U = \lambda_k v_k + U = \lambda_k(v_k + U),$$

where λ_k is the eigenvalue corresponding to the eigenvector v_k. Hence, $\{v_{m+1} + U, \ldots, v_n + U\}$ is a basis of V/U consisting of eigenvectors of T/U, so T/U is diagonalizable. ■

Exercise 5D19

Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there exists a subspace U of V that is invariant under T such that $T|_U$ and T/U are both diagonalizable, then T is diagonalizable.

Axler Note. *See* Exercise 5C13 *for an analogous statement about upper-triangular matrices.*

Solution. This statement is false. Consider the operator $T \in \mathcal{L}(\mathbb{F}^2)$ whose matrix with respect to the standard basis of $\mathbb{F}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Let $U = \operatorname{span}\{(1,0)\}$. Then U is invariant under T since $T(1,0) = (0,0) \in U$. The operator $T|_U$ is diagonalizable because it is the zero operator on a one-dimensional space. The quotient operator V/U is one-dimensional, so T/U is diagonalizable. However, T is not diagonalizable because the only eigenvalue of T is 1, and $\dim E(1,T) = 1 < 2 = \dim \mathbb{F}^2$. ■

Exercise 5D20

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if the dual operator T' is diagonalizable.

Solution. Suppose T is diagonalizable. Then there exists a basis of V consisting of eigenvectors of T. Let $\{v_1, \ldots, v_n\}$ be a basis of V consisting of eigenvectors of T, and let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues. Let $\{\varphi_1, \ldots, \varphi_n\}$ be the dual basis of V' corresponding to $\{v_1, \ldots, v_n\}$. Then for each

$k = 1, \ldots, n$, we have

$$T'(\varphi_k)(v_j) = \varphi_k(T(v_j)) = \varphi_k(\lambda_j v_j) = \lambda_j \varphi_k(v_j) = \begin{cases} \lambda_k & j = k, \\ 0 & j \neq k, \end{cases}$$

so $T'(\varphi_k) = \lambda_k \varphi_k$. Hence, $\{\varphi_1, \ldots, \varphi_n\}$ is a basis of V' consisting of eigenvectors of T', so T' is diagonalizable.

Conversely, suppose T' is diagonalizable. Then there exists a basis of V' consisting of eigenvectors of T'. Using the same reasoning as above, $(T')'$ is diagonalizable with the same eigenvalues as T'. From Exercise 3F32, we know that V is isomorphic to V'' with the map $\Lambda : V \to V''$ defined by

$$\Lambda(v)(\varphi) = \varphi(v)$$

for each $v \in V$ and $\varphi \in V'$. Then $(T')'(\Lambda(v)) = \Lambda(T(v))$ for each $v \in V$, so T has a basis of eigenvectors corresponding to the same eigenvalues as T', so T is diagonalizable. ■

Exercise 5D21

The ***Fibonacci sequence*** $F_0, F_1, F_2, \ldots$ is defined by

$$F_0 = 0, \quad F_1 = 1, \quad \text{and } F_n = F_{n-2} + F_{n-1} \text{ for } n \geq 2.$$

Define $T \in \mathcal{L}(\mathbb{R}^2)$ by $T(x, y) = (y, x + y)$.

(a) Show that $T^n(0, 1) = (F_n, F_{n+1})$ for each nonnegative integer n.

(b) Find the eigenvalues of T.

(c) Find a basis of $\mathbb{R}^2$ consisting of eigenvectors of T.

(d) Use the solution to (c) to compute $T^n(0, 1)$. Conclude that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

for each nonnegative integer n.

(e) Use (d) to conclude that if n is a nonnegative integer, then the Fibonacci number F_n is the integer that is closest to

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

Axler Note. *Each F_n is a nonnegative integer, even though the right side of the formula in (d) does not look like an integer. The number*

$$\frac{1+\sqrt{5}}{2}$$

*is called the **golden ratio**.*

Exercise 5D22

Suppose $T \in \mathcal{L}(V)$ and A is an n-by-n matrix that is the matrix of T with respect to some basis of V. Prove that if

$$|A_{j,j}| > \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}|$$

for each $j \in \{1, \dots, n\}$, then T is invertible. *This exercise states that if the diagonal entries of the matrix of T are large compared to the nondiagonal entries, then T is invertible.*

Exercise 5D23

Suppose the definition of the Gershgorin disks is changed so that the radius of the k^{th} disk is the sum of the absolute values of the entries in column (instead of row) k of A, excluding the diagonal entry. Show that the Gershgorin disk theorem (5.67) still holds with this changed definition.